The Shape of Wisdom: Answers Have Places in Language Models

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Abstract

When a language model answers a factual question, where does its internal state go? We asked four open models (Llama-3.1-8B, Qwen2.5-7B, Gemma-2-9B and Llama-3.2-1B) 2,256 questions whose answers recur, and recorded the state from which each answer is produced at every layer. Questions with the same answer converge on a shared region, the answer's *place*. They do not drift there gradually: in every model the answer becomes readable within a few layers at a fixed depth, between 0.47 and 0.71 of the way through the network, and a single step describes this far better than steady convergence. Places are concepts rather than words. A place learned from people's citizenships finds the country of a film or a town, survives the removal of the answer words' output directions, and is absent in an untrained network. Places are organised by what the model believes, not by what is true, and they are causal: after the step, moving a wrong answer's state into the right place makes the model give the right answer up to 97% of the time, while a random move of the same length works 2%–4% of the time and moves before the step do nothing. Finally, where a wrong answer sits predicts whether the model still recognises the right one in multiple choice better than the model's own confidence does, in all four models.

1 Introduction

Ask a language model "What country is Aníbal Ponce a citizen of?" and, a few dozen layers later, it answers "Argentina". Between the question and the answer the model's internal state at the answering position changes at every layer. This paper asks what shape that change has. Does the state wander, drift slowly toward the answer, or arrive somewhere definite? And if it arrives somewhere, is that somewhere the same for every question with the same answer?

We find that it is. As a question passes through the network, the state moves into a region shared by all questions to which the model gives the same answer. We call this region the answer's *place*. Figure 1 shows it directly: early in Llama-3.1-8B, questions whose answers are twelve different countries form one tangled cloud; within a few layers they separate into one strand per country.

Four properties make places more than a picture. First, they appear suddenly. In each of four models the answer becomes readable from the state within a few layers at a model-specific depth, and a step describes the change far better than gradual convergence (Section 3). Second, a place is a concept, not a word: the same place is reached from different relations and wordings, and it does not depend on the output directions of the answer's own name (Section 4). Third, places are causal. Moving a state into a place after the step makes the model give that place's answer, while random moves of the same size do nothing and moves before the step are ignored (Section 5). Fourth, places are organised by the model's belief, and this makes wrong answers legible: where a wrong answer sits predicts whether the model still recognises the right one, better than the model's confidence does (Section 6).

These results give an intuition a precise form. A model that knows a fact does not hold a vague tendency toward the answer that sharpens layer by layer. It has a definite place for the answer and, at one depth, moves the question there. A confident mistake goes through the same motion to the wrong place: the geometry follows what the model believes, and the truth only decides which place it should have been.

2 Setup

Questions whose answers recur. To see whether questions with the same answer end up in the same internal state, many different questions must share each answer. We built such a set from Wikidata [36]. It contains 2,256 facts with 62 possible answers of four kinds: 24 countries, 16 cities, 12 languages[1] and 10 sports. Each answer is reached through one to three relations. A country is the answer to "which country is this person a citizen of", "which country is this town in" and "which country is this film from"; a city answers questions about birthplaces and company headquarters; a language answers questions about the language a person spoke and the original language of a film; a sport answers "which sport does this athlete play". Subjects were sampled evenly across three popularity bands (by the number of Wikipedia language editions), because models know popular facts far better than rare ones [22]; this gives a mix of facts the models know and facts they do not. Every subject has exactly one value for its relation in Wikidata, a unique English label and no namesake of the same kind.

Three wordings and two formats. Each fact is asked as an open question in up to three wordings: one plain template per relation ("What country is Aníbal Ponce a citizen of?"), and

[1] One relation–answer cell, Hindi as the language a person spoke, was dropped after a manual check found its Wikidata labels unreliable; Hindi remains as the language of films.

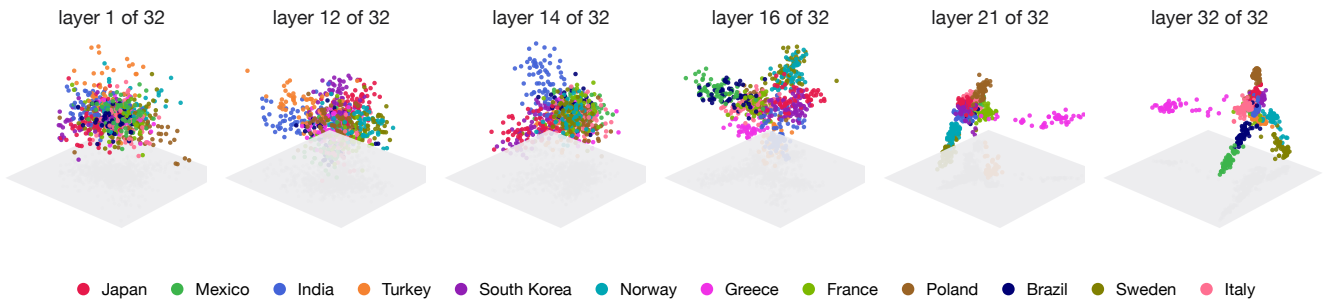


Figure 1: Questions converge on the place of their answer, and they do so at one depth. Held-out questions about citizens, towns and films of twelve countries, as seen by Llama-3.1-8B at six layers. Each panel shows the last prompt position's state in the three directions that best separate the twelve answers among *other* (training) questions at that layer; colour is the country the model answered. Up to layer 14 the questions form one cloud. Answers become readable at layer 15; by layer 16 the questions have begun to separate, and by the last layers each country is a strand of its own. Points are scaled to a robust common range per panel; shadows are drawn on the floor.

two rewordings in assigned styles such as a casual text message, an archivist's catalogue entry or a quiz host's question ("Hi, I'd appreciate knowing which country Aníbal Ponce is a citizen of, if possible?"). The rewordings were written by a language model that was told the subject and what to ask, but never the answer, and were then filtered automatically: each must name the subject exactly, ask a question, mention no candidate answer or its demonym, and differ from the plain wording. 4,323 rewordings passed. The plain question is also asked as four-option multiple choice in all four cyclic orders of the options, and, for a random 30% of facts, with a wrong suggestion from the user ("In which country is Ireneo Portela? I'm fairly sure it's Mexico."), which tempts the model away from an answer it knows. In total each model sees 16,294 prompts, in its own chat format.

Models and what we record. We study Llama-3.1-8B-Instruct [12], Qwen2.5-7B-Instruct [32], Gemma-2-9B-it [7], the smaller Llama-3.2-1B-Instruct, and a randomly initialised copy of Llama-3.1-8B as a control with no learned structure. The model answers greedily. At the last position of the prompt, the position from which the answer is produced, we record the residual stream at every block boundary: its value entering the first block and after each block, before the final normalisation, exactly as the model computes it in bfloat16. A free-form reply is graded only if its first line is one candidate answer (matched against hand-curated names such as "China" or "Mandarin"); replies that hedge, refuse or name two answers are set aside as unclear rather than counted as errors, and a single answer outside the candidate list counts as wrong.

Places. We define an answer's *place* at a given layer as the average state of all questions to which the model gave that answer. Places are defined by what the model said, not by the true answer, so a confident mistake is grouped with the answer it names. Subjects are split in half once. Places, and every other quantity fitted to data, use only training subjects; every reported score uses only the other half. A place needs at least five prompts from at least three training subjects.

Two measurements of convergence. *Readability* asks whether the answer can be read from where the state is: for a held-out question, find the nearest place of the same answer kind and check whether it is the answer the model gave. Distances are measured after a per-layer whitening fitted on training questions (centre, keep 128 principal directions, scale each to unit variance), because a handful of very large directions in the residual stream otherwise dominate Euclidean distance. *Tightness* asks how tightly questions gather around their places, in the model's own unwhitened space: the average squared distance of held-out questions from their place, divided by the spread of the places themselves. Uncertainty comes from resampling subjects. A shuffled control repeats the readability test after permuting answers between facts.

3 Answers arrive in a step

Figure 2a shows how often the answer a model gives can be read from where its state is, layer by layer, for held-out questions. In every model the curve has the same shape: a low plateau, a jump within a few layers, and a high plateau. In Llama-3.1-8B readability rises from below 30% to 71% between layers 13 and 15 and reaches 94%; in Qwen2.5-7B the jump comes later, at layer 20 of 28, and reaches 92%. The depth of the jump differs between models (Table 1) but in each it is a sharp event, not a slow approach.

The early plateau is not zero. Before the jump, 20–55% of answers can be read, more than the shuffled control (6%) and more than the question's text alone (11%–18%). We think this early signal reflects surface cues the model has already picked up from the subject's name, such as the language a name or title is in; we did not test this. Whatever its source, it does not grow with depth, and the jump goes far beyond it.

Three controls show that the structure is learned. An untrained Llama-3.1-8B, given the same prompts and grouped by the trained model's answers, never exceeds 8%. Shuffling answers between facts gives 6%–8% at every layer. A classifier on the question's words alone reaches at most 18%.

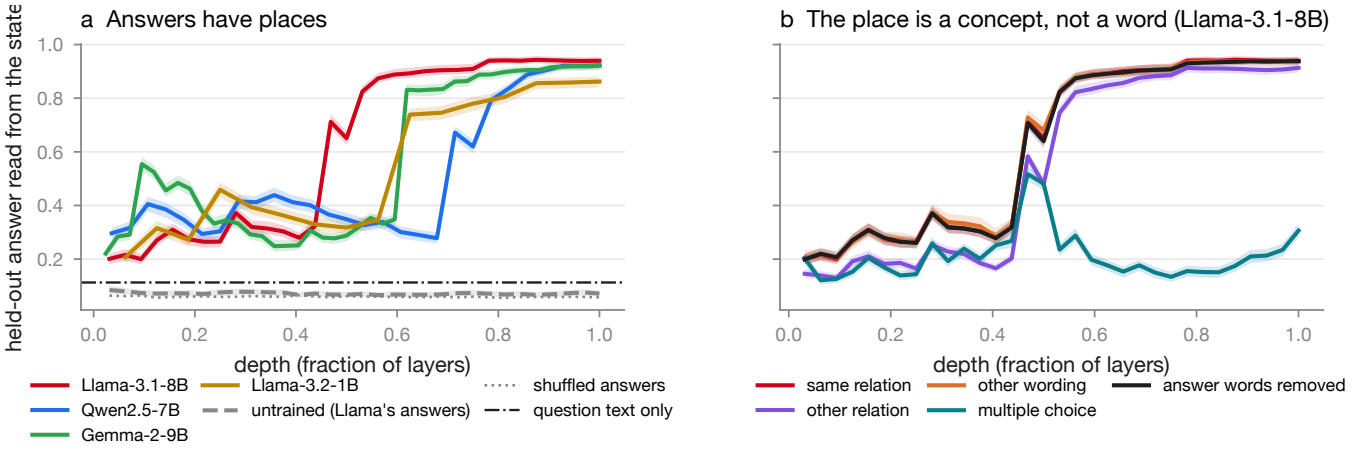


Figure 2: Answers become readable at one depth, and what is read is a concept. Held-out readability: the share of held-out questions whose nearest place (fitted on training subjects, same answer kind) is the answer the model gave; bands are 95% intervals from resampling subjects. (a) All four models, with three controls: the same test on an untrained Llama-3.1-8B using the trained model's answers, labels shuffled between facts, and the question's text alone. (b) Llama-3.1-8B: places fitted on one relation and tested on the others ("other relation"); fitted on plain questions and tested on rewordings ("other wording"); with the answer words' output directions removed; and tested on the states of the multiple-choice version of the same questions.

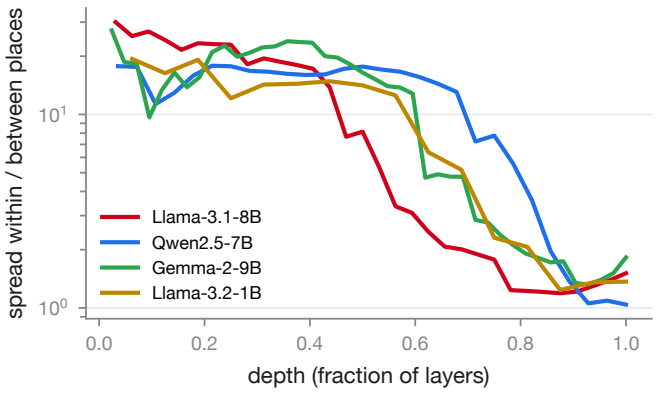


Figure 3: Tightness falls in a step. Average squared distance of held-out questions from their place, divided by the spread of places, in each model's own space (log scale).

Tightness tells the same story. Readability depends on a whitening of the state. The tightness ratio does not: it compares, in the model's own space, how far held-out questions sit from their place with how far places sit from each other. It falls 16- to 25-fold through the network (Figure 3), and most of the fall happens at once. We fitted two shapes to the logarithm of each model's curve, a straight line (the same contraction at every layer, as seen in trained classifiers [17]) and a single smooth step, and compared them by how well each predicts a left-out layer. The step wins in every model, by a factor of 5 to 24 in prediction error (Table 1).

	Llama 8B	Qwen 7B	Gemma 9B	Llama 1B
Layers	32	28	42	16
Accuracy	63%	58%	67%	48%
Readable from (depth)	0.47	0.71	0.62	0.62
Peak readability	94%	92%	92%	86%
Error, step fit	0.025	0.017	0.051	0.034
Error, line fit	0.137	0.422	0.290	0.198
Other relation	91%	85%	90%	76%
Answer words removed	94%	90%	91%	85%

Table 1: Per-model summary. "Readable from": the layer, as a fraction of depth, at which readability first covers half the distance from chance to its peak, chosen on training subjects only. Step and line fits: mean squared error in predicting a left-out layer's log tightness. The last two rows are readability at 90% of depth.

4 A place is a concept, not a word

If a place were only the model preparing to say a word, it would depend on how the question was asked and on the word's output direction. It depends on neither. A place fitted only on questions about people's citizenship finds the right country for questions about towns and films just as well: at 90% of depth, 91% for Llama-3.1-8B, against 94% within the same relation (Figure 2b). Places fitted on plain questions work on casual or formal rewordings (94%). And removing from the state every direction the output layer uses to write any of the 62 answer names, with the final normalisation's scaling taken into account, leaves readability unchanged (94%); removing a random subspace of the same size does the same. The output layer does read these directions, so they are present, but the place does not depend on them.

The multiple-choice version of each question shows where the concept stops. When the options are listed and the model must answer with a letter, the content place is briefly readable from the multiple-choice state at the step (52% in Llama-3.1-8B, against a shuffled level near 6%), then fades to 21% as the state turns toward a letter. This matches the finding that models decide on the content of an option before binding it to its letter [38, 39].

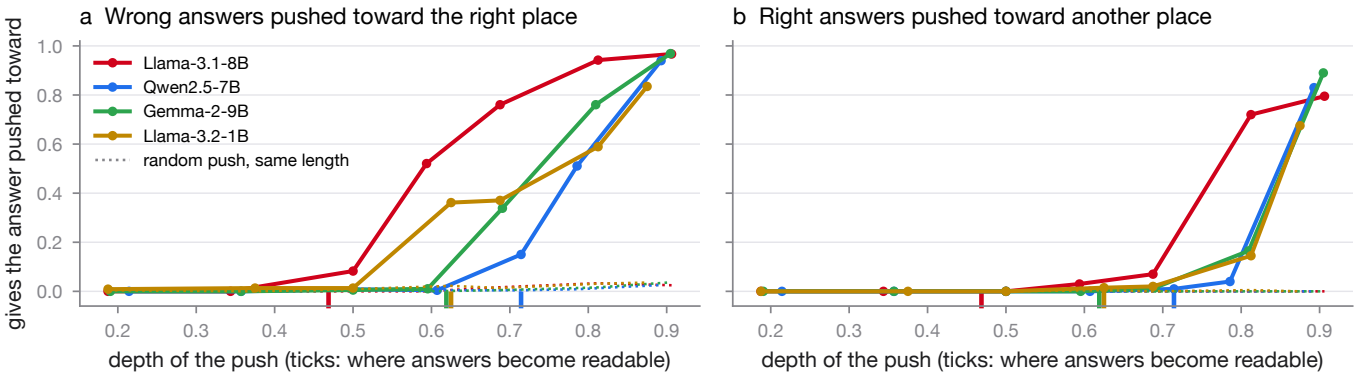


Figure 4: Moving a state into a place changes the answer, but only after the step. (a) Wrong answers moved toward the right answer's place (solid) or by a random direction of the same length (dotted). (b) Right answers moved toward a different answer's place. The y-axis is the share of questions for which the model then gives the answer it was moved toward. Ticks under the axis mark the depth at which each model's answers become readable. Held-out questions; places from training subjects.

5 Moving the state moves the answer

Readability shows that the answer can be read from the place; it does not show that the model uses it. To test this we moved states. For a held-out question that the model answered wrongly, we added to its state, at one layer and at the last prompt position only, the difference between the right answer's place and the place of the answer it gave, and let the model answer again. As controls, we added a random direction of the same length, and we moved the state toward a third answer's place.

The result follows the step (Figure 4). Before a model's answers become readable, moves toward the right place do nothing: at 20% of depth they correct 0%–1% of wrong answers. After the step they work, and they work more the later they are made, reaching 97% (Llama-3.1-8B), 94% (Qwen2.5-7B), 97% (Gemma-2-9B) and 83% (Llama-3.2-1B) at about 90% of depth. Random moves of the same length change 2%–4% of these answers. The effect is not specific to the truth: moved toward a third answer's place, the model gives that answer 83%–95% of the time, and right answers moved toward another place are given up for it 68%–89% of the time. Each model follows the move only from the depth at which its own answers become readable, so the depth of the step measured by reading and the depth at which the model starts to act on the place agree.

Late moves are, in part, close to writing the answer directly, because late states are near the output. The informative parts of the experiment are the depth at which moves start to work, the failure of random moves, and the model's willingness to follow a move to any answer.

6 Where wrong answers sit

Because places are defined by what the model says, a wrong answer has a place too. At the depth where answers become readable, about half of the wrong answers that name a candidate sit nearest the place of the answer they gave (45% for Llama-3.1-8B, 58% for Qwen2.5-7B), as confident mistakes, and 28% of all wrong answers sit nearest the right answer's place. This geometry carries information the output does not. For each wrong free-form answer we asked whether the model still picks the right option in the multiple-choice version (in more than half of the option orders it answered), and predicted this from the model's confidence in its reply, from the geometry of its state (distance to the right place, the right place's rank, and the margin to the nearest place), or from both, with five-fold cross-validation grouped by subject. Geometry predicts recognition better than confidence in every model (Figure 5a): AUC 0.75 against 0.69 for Llama-3.1-8B and 0.73 against 0.50 for Gemma-2-9B, whose confidence carries no signal at all here. Adding geometry to confidence improves AUC in every model; the 95% intervals for the gain, from resampling subjects, are 0.04 to 0.13 (Llama-3.1-8B), 0.08 to 0.19 (Qwen2.5-7B), 0.15 to 0.29 (Gemma-2-9B) and 0.16 to 0.32 (Llama-3.2-1B). The wrong answers studied number 339, 463, 364 and 488.

The hint condition shows the same geometry in a case where we know what happened. For questions a model answers correctly when asked plainly, adding "I'm fairly sure it's . . . " with a wrong country sometimes changes the answer. Where the model resists, its state goes to the right place at the step and stays there (99% of states nearer the right place at the last layer of Llama-3.1-8B). Where the model is misled, the state first goes toward the right place too, with up to 84% of these states nearer it than the suggested place at about half depth, and then leaves it (54% at the last layer; 29% in Qwen2.5-7B). Moving such a misled state back to the right place restores the right answer 93% of the time in Llama-3.1-8B. The model reached the answer it knew, and later layers replaced it.

Places hold beliefs. Two models sometimes give different answers to the same question. If places reflected the question rather than the model's answer, both models' states would sit near the same place. They do not: in all 12 ordered pairs of models, on questions where the two disagree, each model's state is nearer its own answer's place than the other model's answer's place in 66%–88% of cases.

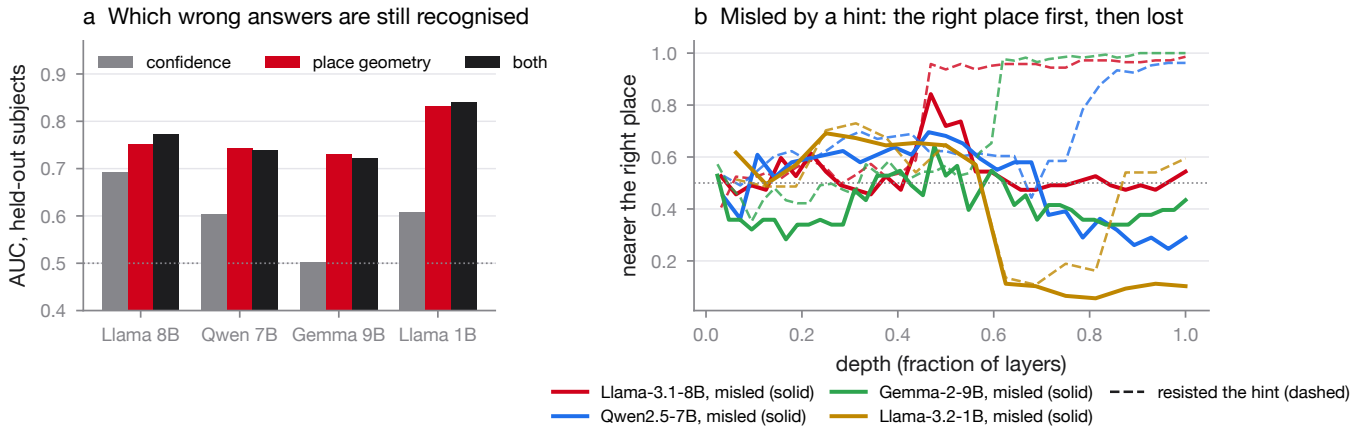


Figure 5: The geometry of a wrong answer says what the model still knows. (a) Among wrong free-form answers on held-out subjects: how well the model's confidence in its reply, the geometry of its state at the step, or both, predict that it still picks the right option in multiple choice (AUC, five-fold cross-validation grouped by subject). (b) Questions answered correctly when asked plainly. With a user's wrong suggestion, those the model resisted (dashed) move to the right place and stay; those it was misled on (solid) first move toward the right place at the step, then away from it. Y-axis: share of states nearer the right answer's place than the suggested answer's place.

7 Related work

Reading intermediate layers. Decoding intermediate states through the output layer, the logit lens [26], and its trained variant, the tuned lens [1], show predictions sharpening with depth; Future Lens [28] and Patchscopes [10] read richer content from a single hidden state. These methods read the output vocabulary. Our places are fitted from the states themselves and survive removing the answer words' output directions, so they capture content that is present before it is expressed.

How depth is used. Several studies find that layers play different roles: early layers gather and detokenise, middle layers build abstract features, and late layers refine a prediction [3, 4, 13, 20, 35]. Correct and incorrect predictions follow similar layer-wise confidence curves [19]. Factual recall first moves relation and then subject information to the last position [9], where feed-forward layers act as key–value memories [8] and specific middle layers are causally decisive [24]. Our step in readability is a population-level, model-independent signature of the moment this recall lands.

Shared concept spaces and representation geometry. Middle layers represent the same content in a shared space across languages and modalities [37, 41]. Concepts have structured geometry: categories form simplices [30], some features are circular [5], space and time are linearly decodable [14], counts lie on curved manifolds that attention reshapes [15], and truth has a linear direction [23]; Goodfire's Neural Geometry series [11] argues that such structure is central to understanding models. In trained classifiers, same-class representations separate from other classes by a roughly constant factor per layer [17], and language models show such class collapse at their top layer [40]; prompts that produce the same next token occupy stable regions [18], and yet no two prompts map to the same state [25]. Places reconcile the last two facts: questions converge in the few directions that carry the answer and stay distinct elsewhere.

What models know but do not say. Internal states often encode more than the output shows [6, 27, 33], but on knowledge benchmarks the gap is small [31], and internal states mostly reflect what the model recalls rather than what is true [2]. Misleading context can overwrite a correct internal answer at a critical layer [16], and users' suggestions shift answers [34]. For multiple choice, the answer letter is produced by a few middle-layer attention heads [38], after the winning option is decided in content space [39], and models carry a prior over letters [42].

Steering. Adding a direction to the residual stream changes behaviour [21, 29]. Our pushes are of this kind, but each one moves a single question's state from one measured place to another, at one layer, and is compared with random pushes of the same length.

8 Limitations

The questions are factual and short, and their answers come from a closed set of 62 countries, cities, languages and sports; places for open-ended answers, multi-step reasoning or long contexts may look different. We studied four instruction-tuned models between 1 and 9 billion parameters, one prompt format each, and greedy answers. Readability uses a whitening of the state; the tightness ratio does not, and it shows the same step, but both are measurements of where states sit, not of the computation that moves them there. The step fit compares two simple shapes; it shows that a sudden change describes the data much better than a constant contraction, not that the change happens within a single layer. Moves at late layers partly write the answer directly, so the strongest causal evidence is the depth at which moves start to work and the failure of random moves. Wikidata answers are a reference, not

ground truth; we removed one unreliable relation–answer pair after a manual check, and a small number of wrong labels surely remain. The rewordings were written by a language model and filtered automatically.

Code and data. Code, the dataset and all derived results: https://github.com/gut-puncture/shape-of-wisdom.

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